§3 The three principles: SM, EU and FR — part 1 of 5

Reference: sec:three_principles, subsec:s3_setup to subsec:sm_foundational

Source: paper/section3_principles.tex, lines 1–990 (the opening of §3, §3.1 conventions and levels of description, §3.2 the self–environment relation and the premises of the principles, §3.3 SM), at commit e6e12cd. The Japanese manuscript is authoritative. Part 1 of §3 follows 02_definition_5.md. Part 2 of §3 follows in 03_principles_2.md.

Opening of §3: the conceptual definition of the three principles

§3 formalizes three constraint axes bearing on the long-term feasibility of an ACS — self-maintenance (SM), environmental utilization (EU) and feasible response (FR) — on top of the self–environment structure and the feasibility structure fixed in §2. All results of §3 are limited to the deterministic, discrete-time framework of Phase 1, and the strength of each claim does not exceed that of the corresponding definition, assumption and proof.

Definition (def:three_principles_concept): conceptual definition of the three principles. In this paper the **three principles** are an arrangement of the feasibility-maintenance constraints that matter when an ACS with a self–environment boundary persists in time under interaction between its internal state and the external environment state. They are arranged as three distinct constraint axes:

- **SM (Self-Maintenance).** The constraint axis concerned with keeping the internal state of the ACS inside the feasible set K.
- **EU (Environmental Utilization).** The constraint axis concerned with the satisfaction, in the given situation, of the required environmental factors that are critical for one-step feasibility.
- **FR (Feasible Response).** The constraint axis concerned with obtaining a feasible response in the situations where one is needed, and with holding, acquiring, recovering and reconstructing the potential for the needed responses. “Feasible” here is fixed to the feasibility of this paper, namely that the next state belongs to K (rem:fr_scope).

The three axes are all meaningful when an ACS with a self–environment relation is described as a structure persisting in time. The paper does not require the time-indexed satisfaction predicates of the three principles to be true at the same time. Their relation to blockage differs by principle.

- If the time-indexed satisfaction predicate EU is false, or if EU is true and the predicate FR is false, then blockage holds in both cases. At a judgment time, blockage divides exclusively into the case where EU is false and the case where EU is true and FR is false (prop:blockage_dichotomy).
- For the predicate SM being false, this paper does not state blockage as a general consequence.
- From the falsity of any of the predicates this paper does not state collapse as a general consequence, and it does not claim that the three principles exhaust the constraints bearing on persistence.

The three principles are not value norms or objective functions imposed on the ACS from outside; in Phase 1 they are descriptive constraint axes adopted to describe temporal persistence under self–environment interaction. The definition is a conceptual arrangement under a collective name, and it states no satisfaction or non-satisfaction of a principle itself. What has a truth value for each principle is the time-indexed satisfaction predicate of def:SM, def:EU and def:FR.

3.1 Conventions and the level of description

Reference: subsec:s3_setup

§3.1 gives, once, the descriptive conventions that later subsections refer to repeatedly, and the domain of application of the time-indexed satisfaction predicates. A list of the main new symbols of §3 is in Appendix app:symbols_s3.

Structural premises of the three principles. The ACS of §2 distinguishes the internal-state type X from the external-environment-state type E_{ext}, with the fixed space E_{ext} as background type, and describes the interaction of self and environment by the state transition $f : X \times U \times E_{\mathrm{ext}} \to X$. The required environment space E and the required-environment satisfaction set $E_{\mathrm{req}}(x)$ describe that the persistence of the structural self is not independent of external environmental conditions. Taking this self–environment structure as given, §3 formalizes three problems as SM, EU and FR: the current feasibility; the satisfaction of the required environmental factors that are critical for one-step feasibility; and a feasible response in the situations where one is needed. Phase 1 does not treat the generation of the self–environment boundary, ontogenesis, or the physical mechanism of boundary formation.

For the evaluation-related quantities introduced in §3, the global convention of subsec:description_levels_global applies: the structural evaluation function needed on the ACS side is distinguished from the mathematical formal representation used to describe it in the paper. This paper does not state, as a general consequence of the existence of a formal quantity, that the ACS knows its true value in full or implements the same formula internally.

Conventions common to §3. Six remarks fix the conventions.

- **Treatment of implication (rem:conv_nonimplication).** §3 does not state well-formed-realization failure $\mathrm{WF}_{i,t} = 0$ or collapse $\mathrm{Col}_{i,t} = 1$ as a general consequence of any time-indexed satisfaction predicate being false, or of T-STAB not being satisfied. For blockage $A_t = \emptyset$: (1) the falsity of SM_t, the non-satisfaction of T-STAB, and the holding of $\chi_{\mathrm{EI},t} = 0$ at a single time do not by themselves give blockage as a general consequence; (2) if EU_t is false then $A_t = \emptyset$ (prop:eu_blockage); (3) if FR_t is false then $A_t = \emptyset$ (def:FR). For the relation quality Q, the sign (positive, zero, negative) when it is defined is distinguished from the case where it cannot be evaluated at all. A table showing the time domain $\mathbb{T}_{\mathrm{P},i}$ explicitly is in rem:principles_failure_table.
- **Level-of-description convention (rem:conv_description_level).** The formal quantities introduced in §3 are theoretical description under subsec:description_levels_global and do not require the ACS to hold or compute their true values or whole sets internally. What internal evaluation and response use is the information actually available to that ACS. The non-degeneracy required of the structural evaluation function is stated in subsec:self_env_principles.
- **Non-normativity (rem:conv_nonnormative).** The three principles and the evaluation quantities of §3 are descriptive constraint axes and formal representations bearing on long-term feasibility. They are not value norms, a single objective function or a maximization target. This does not require identifying the truth of a predicate with moral good, nor regarding Q or T-STAB as a single utility function.
- **Non-exhaustiveness and non-minimality (rem:conv_nonexhaustive).** §3 does not claim that the three principles exhaust all constraints bearing on persistence, that the three-way decomposition is unique, that the three predicates are mutually non-redundant, or that the set of principles is minimal. The relations among the principles are stated in subsec:principle_linkage as linkage and a limitation of implication relations.
- **Distinction of levels: coupling, realized interaction, EU, relation evaluation (rem:conv_ro_layers).** §3 distinguishes the following mathematical objects. $\mathrm{EC}(x_t)$ is the structural environment coupling condition of def:env_coupling; $\chi_{\mathrm{EI},t}$ is the binary indicator of realized external interaction of def:phi; $e_t \in E_{\mathrm{req}}(x_t)$ is the proposition expressing the current required-environment satisfaction; EU_t is the time-indexed satisfaction predicate of def:EU; $Q_{i \leftarrow j,t}$ is the relation quality of §6 (sec:rel_min), which has a value only on its domain. They are not identified, and where equivalence or implication is used, it rests on an individually stated condition. The generalized-gradient sensitivity of prop:pi_E_sensitivity is a local feasibility-sensitivity condition separate from the structural environment coupling condition.

- **Status of T-STAB (`rem:conv_tstab_scope`).** T-STAB is a finite-window time-integration condition on principle-related evaluation. It is none of: Lyapunov stability, asymptotic stability or structural stability in dynamical-systems theory; a fourth principle added to the three; an exogenous objective function. This paper does not state, as a general consequence of its satisfaction alone, infinite-time feasibility, convergence, attractivity, existence of an invariant set or arrival at a fixed point.

Levels of description and the domain of the predicates

`Reference: subsubsec:s3_levels`

The levels of description in §3 (`rem:s3_description_levels`). §3 distinguishes the following levels.

- *Principle.* SM, EU and FR are constraint axes on a persisting ACS. §3 states no satisfaction or non-satisfaction of a principle itself.
- *Principle response item.* For each principle, an item stating what is asked to hold on the ACS side. It does not name means.
- *Time-indexed satisfaction predicate.* The predicates SM_t, EU_t, FR_t defined for each principle and each time. Truth and falsity belong to this level. §3 writes "true" and "false"; "false" means that the predicate is defined and is false. A time at which the predicate is undefined is not counted among the times at which it is false.
- *Structural functions and standard functional forms.* The basic functions and base functions of §2, the principle-related structural functions (classification A) of §3, and TIPA, PSL and ISL of §4 (classification B). Operation, stoppage and degradation are stated of this level.
- *Diagnostic quantities.* The response diversity $\mu_{\mathrm{eff},t}$, the robust persistence horizon, the evaluation scales $s_{\cdot,t}$ and other classification-C quantities.

This paper does not state, as a general consequence, that a function operating implies a predicate true, or that a predicate false implies the stoppage of a particular function. No function dedicated to a single principle is posited. That several structural functions jointly bear one principle response item, and that one structural function takes part in several items, are not excluded.

Separately from these levels, §3 distinguishes the theoretical quantities of §3 (Π, A_t, $\mathrm{Crit}^{(1)}(x_t)$ and so on), the ACS's internal estimation and evaluation concerning them, and measurement and diagnosis by an external observer. The truth condition of a time-indexed satisfaction predicate is fixed by the theoretical quantities alone, and uses neither the result of internal estimation or evaluation nor the result of an external observer's measurement (`rem:crit_description_level`).

§3 treats a single ACS individual Σ_i. Following the convention of §2, the individual index is omitted on the state, environment, feasible action set and the like, and these are written x_t, e_t, A_t. The indicators bearing on identity of the individual, $\mathrm{WF}_{i,t}$, $\mathrm{Col}_{i,t}$, $\mathrm{FT}_{i,t}$, keep the index.

Definition (`def:principle_time_domain`): principle time domain. For an ACS individual Σ_i,

$$\mathbb{T}_{\mathrm{P},i} := \{t \in \mathbb{N}_0 \mid \mathrm{FT}_{i,t} = 0\}$$

is called the **principle time domain**, where $\mathrm{FT}_{i,t}$ is the functional-termination indicator of `def:functional_termination`. The time-indexed satisfaction predicates of §3 are defined only at times $t \in \mathbb{T}_{\mathrm{P},i}$ (§2, `rem:collapse_not_principle_failure`). At a time $t \notin \mathbb{T}_{\mathrm{P},i}$, no time-indexed satisfaction predicate is made true or false.

Lemma (`lem:principle_time_domain_down`): downward closedness of the principle time domain. If $t \in \mathbb{T}_{\mathrm{P},i}$, $s \in \mathbb{N}_0$ and $s \le t$, then $s \in \mathbb{T}_{\mathrm{P},i}$. Moreover, if $t \in \mathbb{T}_{\mathrm{P},i}$ then $\mathrm{Col}_{i,t} = 0$.
Proof outline. If $s < t$ and $\mathrm{FT}_{i,s} = 1$, then by (C-2) of `prop:collapse_basic` $\mathrm{FT}_{i,t} = 1$, contradicting $t \in \mathbb{T}_{\mathrm{P},i}$; the case $s = t$ is the hypothesis. The second statement is the contrapositive of $\mathrm{Col}_{i,t} = 1 \Rightarrow \mathrm{FT}_{i,t} = 1$ from `prop:termination_order`(ii).

The principle time domain, well-formed realization and trajectory records (`rem:principle_time_domain_scope`). The principle time domain does not impose $\mathrm{WF}_{i,t} = 1$. §2 takes $x_t \in X$ and $e_t \in E_{\mathrm{ext}}$ at any time of a realized state and assumes no implication in either direction between functional termination and well-formed realization (`rem:collapse_wf_relation`). Hence times at which $\mathrm{WF}_{i,t} = 0$ holds temporarily before functional termination are also objects of the judgment of the predicates. Likewise the domain does not impose $x_t \in K$. Times at which the state is outside the feasible set but functional termination has not been reached — for example, a living body whose state has fallen below a threshold but is on its way to recovery, or a robot whose power is below a threshold but which retains a means of return — are objects of judgment (the status of examples is governed by `rem:example_status`). At such times SM_t is false, and EU_t and FR_t describe the situation concerning recovery to K in one step (`subsec:eu`, `subsec:fr`). By `lem:principle_time_domain_down`, if $t \in \mathbb{T}_{\mathrm{P},i}$, then for a trajectory record σ recording the sequence of collapse indicators of Σ_i we have $t \in I_{\mathrm{ACS}}(\sigma)$ (`def:acs_valid_time_domain`, §7). Conversely, $I_{\mathrm{ACS}}(\sigma)$ can contain times after functional termination, so it is not used as the principle time domain.

3.2 The self–environment relation and the premises of the principles

Reference: `subsec:self_env_principles`

§3.2 arranges, from the self–environment structure established in §2, why the three principles are introduced in this theory, and gives the premises of each and a unified descriptive form. The formal definitions of the predicates are in `subsec:sm_foundational`, `subsec:eu` and `subsec:fr`. §3.2 places no new assumption, proposition or proof.

Premises on the self-environment boundary and interaction. §2 describes the ACS as $\Sigma = (X, U, E, f, \Pi, M, \mathrm{Update})$ with the fixed space E_{ext} as background type, and under the type-separation convention TSC distinguishes X from E_{ext}; this is described as the self–environment boundary $B_{SE} = (X, E_{\mathrm{ext}})$ (`def:b_se`). The point is not to generate one of X and E_{ext} from the other over time. Phase 1 does not treat the temporal development of boundary formation and takes as given a formal structure in which inside and outside are distinguished relative to the boundary. Accordingly the "simultaneous establishment" of self and environment in this paper means *structural simultaneity*: that the two are distinguished relatively by the same boundary relation.

The boundary does not mean a severing of self and environment. The state transition map $f : X \times U \times E_{\mathrm{ext}} \to X$ (`def:state_transition`) describes the next state as a function of the current state, the effective action and the realized external environment state, so the external environment state enters the temporal development of the ACS as a formal variable that can take part in the transition. The paper does not, however, assume from the type of the map non-degeneracy in the environment argument of the kind $e \neq e' \Rightarrow f(x,u,e) \neq f(x,u,e')$. That different external environment states give the same next state for the same (x,u) is formally allowed. What this paper requires as dependence on required environment is given not by the argument type of f but by the required environment space E and the constitutive requirement CR-1 (`subsec:environment_space`).

The required environment space E describes, state-dependently and under the fixed E_{ext}, the environmental factors needed for the ACS's persistence. The required-environment satisfaction set $E_{\mathrm{req}}(x) = \{e \in E_{\mathrm{ext}} \mid \forall \eta \in E,\ \eta(x,e) = 1\}$ (`def:E_req`) represents the external environment states that satisfy all required environmental factors at state x. Thus the structural self of an ACS is distinguished from the external environment, while part of its persistence conditions can depend on external environmental conditions.

The persistence evaluation function Π, the feasible set K and the feasible action correspondence $A(x,e)$ (`subsec:viability_function_and_blockage`) describe whether, under the current state and the realized external environment state, a one-step state transition through the effective action can remain in K. "Through an action" does not mean directed intervention only: it means describing the one-step feasibility-maintenance possibility through the elements of the action space U of this paper, including selective waiting. So the ACS of Phase 1 is described as a structure that distinguishes inside from outside, makes state transitions with the external environment state as an argument, and can depend for its persistence on required

environmental conditions. The three principles are introduced on the premise of this self–environment relation.

Three structural questions and the three principles. When an ACS with a self–environment boundary and dependence on required environment is considered as a structure persisting in time, at least three questions are meaningful at once.

- *Question 1 (feasibility of the structural self).* Does the current structural self satisfy the feasibility condition at the current time? Formally this connects to whether the current state belongs to K, that is, whether $\Pi(x_t) \geq 0$.
- *Question 2 (satisfaction of critical required factors).* In a situation where no action can place the next state in K unless a certain required environmental factor is satisfied, is that factor actually satisfied? Such a factor is called a critical required factor (`def:eu_fire`). The situation can arise both when the current state is inside K (maintenance) and outside it (recovery).
- *Question 3 (feasible response in the needed situation).* In a situation where all critical required factors are satisfied, does a response exist that places the next state in K — a response maintaining feasibility, or one recovering from outside K in one step? And is the possibility of such a response held, acquired, recovered and reconstructed over time?

The paper adopts SM, EU and FR as the constraint axes corresponding to these three questions; the predicates are given in `def:SM`, `def:EU`, `def:FR`. The paper does not claim that the three principles are "derived from" the self–environment relation. What it claims is limited to this: when self and environment are distinguished and persistence in time is treated under their interaction and dependence on required environment, three structural problems — the current feasibility, the satisfaction of critical required factors, and a feasible response in the needed situation — become objects of description at the same time, and Phase 1 formalizes them as the three axes SM, EU and FR.

That the ACS is not functionally closed to differences of the external environment state — the structural coupling of self and environment — is none of the three principles; it is treated as a premise and background structure of EU in `subsec:env_coupling`. The formula $\Pi(x_t) \geq 0$ of Question 1 can itself be described formally once X and Π are given, so the paper does not claim that "without a self–environment boundary the formula of feasibility itself is undefinable". The claim is only that, when the three questions are placed together as the persistence problem of one and the same ACS, the self–environment boundary and its interaction structure are a common premise.

Structural simultaneity of the three principles (`rem:principles_simultaneity`). The simultaneity here does not mean that SM, EU and FR occur in sequence at different points on the time axis; it means that, when the persistence of an ACS with a self–environment relation is described, the three constraint problems are logically meaningful side by side. Hence the structural simultaneity of the three principles is not identified with the time-indexed satisfaction predicates of the three being true at all times. That one or more predicates are false at some time is not excluded by the theory. Nor does the paper state $\mathrm{Col}_{i,t} = 1$ as a general consequence of one or more predicates being false; the definition of collapse is in §2 (`def:collapse`) and the arrangement of surrounding states and paths is the responsibility of §7 (`sec:collapse_cond`). The state $A_t = \emptyset$ (blockage, `def:blockage_sec2`) is likewise not identified with collapse or with the failure of well-formed realization. The relations among the predicates are stated in `subsec:principle_linkage` as linkage and a limitation of implication relations.

Premises of the three principles and the unified descriptive form. The premises of the principles are not principles (constraints on a structure) but a description of the class of ACSs and the world under consideration. Each is shown with the kind (convention, constitutive requirement, assumption) it belongs to.

- *Premises of SM.* (P-SM1) The structural self has the feasible set $K = \{x \mid \Pi(x) \geq 0\}$ (`def:Pi`, `def:K`). (P-SM2) The state can change independently of the selection of an action (f_{auto} of `ass:f_decomposition`).
- *Premises of EU.* (P-EU1) The ACS is within an external environment (E_{ext}, convention TSC, B_{SE}, $f : X \times U \times E_{\mathrm{ext}} \to X$). (P-EU2) Environmental input is non-degenerate (`ass:env_input_nondegeneracy`, `ass:env_viable_nondegeneracy`). (P-EU3) There are required environmental factors, and without them the ACS cannot persist (constitutive requirement

CR-1, subsec:environment_space). The structural coupling of self and environment (subsec:env_coupling) is the background structure of EU.

- *Premise of FR.* (P-FR1) The future state is determined depending on the current action and the sequence of realized external environment states (def:state_transition).

ass:nonstationary (non-stationary environment) is not an antecedent of the time-indexed satisfaction predicate FR; it limits the class of objects as a background assumption of Phase 1 (subsubsec:real_world_scope). ass:theta_exterior_blockage is not a premise; it is an additional assumption used only in the conditional analysis of subsec:sd_necessity, in the corollary that connects persistence outside the feasibility kernel to blockage.

The three principles are described in the following unified form. The "principle response item" states what is asked to hold on the ACS side and does not name the means that bear it; the "time range" row is explanatory and not a formal classification. The manuscript gives this as a table; its content is as follows.

Item	SM	EU	FR
Object	state of the structural self	relation between the self and the required environmental factors	response in the needed situation
Domain of application	$t \in \mathbb{T}_{\mathrm{P},i}$	$t \in \mathbb{T}_{\mathrm{P},i}$	$t \in \mathbb{T}_{\mathrm{P},i}$ and $(x_t, e_t) \in \mathcal{D}_{\mathrm{FR}}$ (times at which EU_t is true)
Situation asked about	all judgment times	all judgment times (vacuously true at times where $\mathrm{Crit}^{(1)}(x_t) = \emptyset$)	all times of the domain of application (maintenance inside K; one-step recovery outside it)
Time-indexed satisfaction predicate	$x_t \in K$ (def:SM)	$\forall \eta \in \mathrm{Crit}^{(1)}(x_t) : \eta(x_t, e_t) = 1$ (def:EU)	$A_t \neq \emptyset$ (def:FR)
Principle response item	the current state is inside K, and a state that does not impair future feasibility is formed and held	for the required environmental factors critical to one-step feasibility, the relation of self and environment leading to their satisfaction holds	in the situation $\mathcal{D}_{\mathrm{FR}}$ a response leading to the K side exists, and the potential for the needed responses is held, acquired, recovered and reconstructed over time
Time range (explanatory)	immediate: the current state. Extended: Hor_f, T-STAB (T1)	immediate: satisfaction of the one-step critical required factors. Extended: lasting satisfaction (treated on the CR-1 side)	immediate: local FR. Extended: subsec:sd_necessity, T-STAB (T2)
Relation to blockage	falsity is not a defining condition of blockage	falsity implies blockage (prop:eu_blockage)	falsity coincides with blockage (on $\mathcal{D}_{\mathrm{FR}}$). Blockage divides exclusively into EU false, or EU true and FR false (prop:blockage_dichotomy)

The three principles and the intrinsic evaluation function

The three principles are constraint axes, but differences bearing on them are not to be entirely cut off from the operation of the ACS. As a constitutive requirement on evaluation, (CR-2b)(iii) of CR-2 in §2 requires that there are cases in which different objects of evaluation are treated as non-equivalent (non-degeneracy), and that, when $K \neq \emptyset$ and $X \setminus K \neq \emptyset$, there are cases in which evaluation differs between a state belonging to K and one not belonging to it (feasibility sensitivity) (`def:three_primitives`, `subsec:three_primitive_functions`). §3 does not change the content of the constitutive requirement.

In addition, to describe the functional structure of an ACS that maintains the three principle constraints within self–environment interaction, §3 states, as a functional requirement on the structural evaluation function, that at least part of the differences bearing on the decline of feasibility, the non-satisfaction of required environmental factors, the degeneration of feasible responses, the worsening of external relations and the like is available for forming the ACS's response. When state differences, relation differences or temporal changes bearing on the three principles exist among the objects of evaluation, it is the lowest line of non-degeneracy of the evaluation function that the evaluation result or the internal response can change in response to at least part of those differences. This functional requirement is not a condition of the time-indexed satisfaction predicates, nor a constitutive requirement, nor an assumption; it is a descriptive requirement on the intrinsic evaluation function bearing on the three principles. The paper does not require from it the identification of all distinct states, agreement between formal values and internal evaluation, or a general error bound. Where a quantitative accuracy condition is needed, it is stated explicitly as an additional assumption in the relevant later section.

The paper treats this role as the **structural evaluation function**. It is not a pass–fail convention set arbitrarily by an external observer for the ACS; it is positioned as an evaluation function that the ACS can use in forming responses, based on the information available from self–environment interaction. This intrinsic character does not mean that the ACS recognizes or computes the true values of the formal model in full. For example, for Π, K, $A(x, e)$, $E_{\mathrm{req}}(x)$, $\mathrm{Crit}^{(1)}(x)$ and μ_{eff} used in later subsections, the paper does not require the ACS to represent their true values or whole sets internally in full. Analysis objects derived from the formal model, such as the feasibility kernel, are distinguished from the ACS's intrinsic evaluation function itself. In the later evaluation of feasibility-related differences accompanying external relations and in the temporal integrated evaluation of the three principles, the evaluation function itself is treated as a function intrinsically needed on the ACS side, while concrete scales, thresholds, accuracy, encodings and time windows are not fixed as a particular implementation.

Status of examples (`rem:example_status`). In §3 and later sections, a few intuitive examples concerning humans, other organisms or artificial cognitive systems may be given as supplements, to help intuitive understanding of the structural correspondence of the evaluation function. These examples are not existence proofs of that function, concrete implementations common to ACSs in general, or demonstrations of a causal mechanism. Where an example is stated as an empirical fact, a literature basis is attached; an example without one is not a formal consequence of Phase 1 but an interpretive supplement to help theoretical understanding.

Connection to later subsections. Taking the self–environment boundary, state transition, dependence on required environment and the feasibility structure as premises, SM, EU and FR are introduced as three constraint axes that are meaningful together. `subsec:sm_foundational` treats SM; `subsec:viab_core` the E_{req} robust viability kernel and the finite-step robust feasible sets; `subsec:env_coupling` the structural environment coupling and its auxiliary descriptions and diagnostic quantities; `subsec:eu` EU; `subsec:fr` FR. Then `subsec:principle_failure_relation` arranges the relation of the principles to blockage, well-formed realization and collapse; `subsec:principle_linkage` the linkage of the principles and the limitation of implication relations; `subsec:sd_necessity` the conditional connection between response-diversity degeneration and long-term feasibility; `subsec:s3_tstab` the temporal integrated evaluation of the three principles; and `subsec:principles_to_tipa` the connection to structural functions and standard functional forms. In each subsection, claims are distinguished as definitional equivalences, conditional implications under additional assumptions, or structural connections, so that the

strength of a claim does not exceed that of the proofs and assumptions.

3.3 SM

Reference: subsec:sm_foundational

§3.3 gives the time-indexed satisfaction predicate of SM and the structure of the feasibility margin Π used in its judgment. SM asks whether the state x_t at time t belongs to the feasible set K defined in §2. It is not a new objective function, an optimization rule, or a temporal generating condition that precedes the other two principles. The subsection first arranges the generalized-gradient structure of Π and the componentwise diagnostic quantity that the minimum of Π discards.

The Clarke generalized-gradient structure of Π

Π of def:Pi is defined by the formula $\min_k(x_k - \theta_k)$, which makes sense on all of $\mathbb{R}^n$. Since the Clarke generalized gradient is standardly treated for locally Lipschitz functions on open sets, the paper first defines the extension $\widetilde{\Pi} : \mathbb{R}^n \to \mathbb{R}$, $\widetilde{\Pi}(x) := \min_k(x_k - \theta_k)$, and on the state space X of the ACS uses its restriction $\Pi := \widetilde{\Pi}|_X$. The Clarke generalized gradient is taken for $\widetilde{\Pi}$ and written $\partial^C\widetilde{\Pi}(x)$; for $x \in X$ the abbreviation $\partial\Pi(x) := \partial^C\widetilde{\Pi}(x)$ is used throughout. With this extension the generalized gradient at a boundary point $x \in \partial X$ is consistently defined as that of a locally Lipschitz function on an open neighbourhood in $\mathbb{R}^n$.

Proposition (prop:pi_subdiff): Lipschitz continuity of Π and its Clarke generalized-gradient structure. For the persistence evaluation function $\Pi = \widetilde{\Pi}|_X$ of def:Pi:

- **(i) Lipschitz continuity.** Π is Lipschitz continuous on X with constant $L = 1$: $|\Pi(x) - \Pi(x')| \le \|x - x'\|_\infty$ for all $x, x' \in X$.
- **(ii) Explicit form of the Clarke generalized gradient.** At any $x_t \in X$ the Clarke generalized gradient $\partial\Pi(x_t) = \partial^C\widetilde{\Pi}(x_t)$ is a nonempty compact convex set, and

$$\partial\Pi(x_t) = \mathrm{conv}\{e_k \mid k \in \arg\min_j(x_{j,t} - \theta_j)\},$$

  where $e_k \in \mathbb{R}^n$ is the standard basis vector with k-th component 1 and the others 0.

Proof outline. (i) Each affine function $\ell_k(x) = x_k - \theta_k$ is Lipschitz with constant 1, and the pointwise minimum of finitely many Lipschitz functions is Lipschitz with the same constant, a standard fact that can be checked directly; for the general background of variational analysis the paper refers to Rockafellar and Wets (cited as Rockafellar and Wets 1998). (ii) $\widetilde{\Pi}$ is the pointwise minimum of finitely many C^1 affine functions, so by the standard calculus of the Clarke generalized gradient (cited as Clarke 1990) it coincides with the convex hull of the gradients of the indices attaining the minimum; since $\nabla\ell_k = e_k$ the stated form follows, and the convex hull of finitely many vectors is a nonempty compact convex set.

The L_∞ norm (rem:linf). The L_∞ norm adopted in prop:pi_subdiff(i) is structurally consistent with $\Pi(x_t) = \min_k(x_{k,t} - \theta_k)$ being defined as a minimum over components: the maximum error of each component directly bounds the variation of the feasibility margin Π.

The threshold-violation set and the deficiency vector

Definition (def:violation_profile): threshold-violation set and deficiency vector (classification C). For a state x_t, the **threshold-violation set** is $V^-(x_t) := \{k \in \{1, \dots, n\} \mid x_{k,t} - \theta_k < 0\}$, and the **deficiency vector** is $d^-(x_t) := ((\theta_1 - x_{1,t})_+, \dots, (\theta_n - x_{n,t})_+) \in \mathbb{R}^n$ with $(a)_+ := \max(a, 0)$. $V^-(x_t)$ is the index set of components below threshold and $d^-(x_t)$ the shortfall of each component; $V^-(x_t) = \{k \mid d^-_k(x_t) > 0\}$ holds.

Proposition (prop:pi_violation_profile): relation of Π to the threshold-violation set. $\Pi(x_t) \geq 0 \iff V^-(x_t) = \emptyset \iff d^-(x_t) = \mathbf{0}$, and $\Pi(x_t) < 0 \iff V^-(x_t) \neq \emptyset$. *Proof outline.* $\Pi(x_t) \geq 0$ means $x_{k,t} - \theta_k \geq 0$ for all k, which is $V^-(x_t) = \emptyset$ and also $(\theta_k - x_{k,t})_+ = 0$ for all k; the second equivalence is the negation of the first.

The time-indexed satisfaction predicate of SM and the feasible set

Definition (def:SM): time-indexed satisfaction predicate of SM. For $t \in \mathbb{T}_{\mathrm{P},i}$, SM_t is true when $x_t \in K$ and false when $x_t \notin K$.

Proposition (prop:sm_K_equiv): equivalence of SM and the feasible set. (i) For any $t \in \mathbb{N}_0$, $\Pi(x_t) \geq 0 \iff x_t \in K$. (ii) If $t \in \mathbb{T}_{\mathrm{P},i}$, then SM_t being true, $\Pi(x_t) \geq 0$ and $x_t \in K$ are equivalent. *Proof outline.* (i) is the definition of K (def:K) with $x_t \in X$; (ii) follows from def:SM and (i).

Proposition (prop:sm_one_step_A): one-step retention of K and A_t. (i) For any t and $u \in U$, $u \in A_t \iff f(x_t, u, e_t) \in K \iff \Pi(f(x_t, u, e_t)) \geq 0$. (ii) For any t, $x_{t+1} \in K \iff u_t \in A_t$. (iii) If $t+1 \in \mathbb{T}_{\mathrm{P},i}$, then SM_{t+1} is true if and only if $u_t \in A_t$. *Proof outline.* (i) is the definition of the feasible action correspondence (def:at) and of K; (ii) follows from $x_{t+1} = f(x_t, u_t, e_t)$ (def:state_transition) and (i); (iii) from (ii) and def:SM.

SM, blockage, collapse and required-environment satisfaction distinguished

The foundational position of SM (rem:sm_foundational_status). In the sense shown by prop:sm_K_equiv, SM is the constraint axis among the three that directly describes whether the current state belongs to the feasible set K. "Foundational" here refers to a position in the formal system: K is referred to not only for the judgment of the current state but also as the reference set for the feasible action set A_t, the one-step critical region $\mathrm{Crit}_\eta^{(1)}$ of EU, the domain $\mathcal{D}_{\mathrm{FR}}$ of FR, the feasibility-retention indicator function φ_u and the response diversity $\mu_{\mathrm{eff},t}$. $\mathrm{Crit}_\eta^{(1)}$ and $\mathcal{D}_{\mathrm{FR}}$ refer to K through A, but the domains of EU_t and FR_t do not require $x_t \in K$ (def:EU, def:FR). Hence this paper does not state the truth of EU_t or FR_t as a general consequence of the truth of SM_t, nor claim that SM logically contains the other two principles (rem:principle_implication_scope).

The current SM and next-time retention of K are separate conditions (rem:sm_current_next). prop:sm_one_step_A distinguishes the state condition at the current time from the possibility of retaining K one step ahead. The SM of the current time is the state condition $x_t \in K$; $A_t \neq \emptyset$ is a transition condition: under the realized external environment state e_t, an action placing the next step in K exists. So the paper does not assume $x_t \in K \iff A_t \neq \emptyset$ from the definitions alone. The treatment of times at which $A_t = \emptyset$, both for $x_t \in K$ and for $x_t \notin K$, is arranged in the dichotomy of blockage (prop:blockage_dichotomy) in subsec:principle_failure_relation.

SM and blockage are logically distinct (rem:sm_blockage_distinction). $x_t \in K$ and blockage $A_t = \emptyset$ are not the same condition. SM is fixed by the membership of the current state x_t in K, while A_t is fixed by the existence of an action u with $f(x_t, u, e_t) \in K$: the former is a state condition at time t, the latter a transition condition expressing whether an action reaching a feasible state one step ahead exists from the same state and realized environment. The general definitions of Phase 1 assume neither $A_t = \emptyset \Rightarrow x_t \notin K$ nor $x_t \notin K \Rightarrow A_t = \emptyset$. Whether one of the implications holds in an individual ACS depends on the concrete structure of f, U, e_t and K.

The falsity of the SM predicate is distinguished from collapse (rem:sm_not_collapse). That SM_t is false at $t \in \mathbb{T}_{\mathrm{P},i}$, that is, $\Pi(x_t) < 0$ ($x_t \notin K$), expresses that the state at time t is outside the feasible set. This is not the same as collapse $\mathrm{Col}_{i,t} = 1$ of §2 (def:collapse). The paper does not assume $x_t \notin K \Rightarrow \mathrm{Col}_{i,t} = 1$ in general. At a time at which SM_t is judged, $\mathrm{Col}_{i,t} = 0$ by lem:principle_time_domain_down. Conversely, the paper does not build the judgment condition of collapse into the definition of SM: SM is a single-time state condition, and collapse is a concept of §2 that includes non-recoverability as the same individual (def:collapse). For the failure of well-formed realization $\mathrm{WF}_{i,t} = 0$ as well, this paper does not state it as a general consequence of the falsity of SM_t alone (rem:principles_failure_table).

SM and required-environment satisfaction are distinguished (`rem:sm_vs_Ereq`). The required-environment satisfaction set $E_{\mathrm{req}}(x_t)$ represents, relative to the state x_t, the external environment states satisfying all required environmental factors, whereas SM asks whether the current state x_t itself belongs to K. So $e_t \in E_{\mathrm{req}}(x_t)$ and $x_t \in K$ are separate conditions, and the paper does not assume their equivalence. This does not mean that the satisfaction of required factors is irrelevant to the persistence of the structural self: CR-1 requires that the persistence of the structural self is related to external environmental conditions (`subsec:environment_space`). But the relation is not expressed as a single-time equivalence of the form "if the required factors are currently satisfied, the current state belongs to K". Internal damage, the state of internal resources, history and other state differences can be contained in x_t, so the general definition does not exclude the case where the required factors are currently satisfied and yet $x_t \notin K$. Nor does it assume that a temporary non-satisfaction of required factors immediately yields $x_t \notin K$ or collapse. The satisfaction of those required factors that are critical at time t is treated as the time-indexed satisfaction predicate of EU (`subsec:eu`).

Required-environment dependence and the structural connection of SM (`rem:sm_environment_dependency`). SM is judged by $\Pi : X \to \mathbb{R}$, which takes the internal state only as argument, but this does not mean that SM is causally or structurally independent of the external environment. The state transition $f : X \times U \times E_{\mathrm{ext}} \to X$ has the external environment state as an argument, and the required environment space E and CR-1 describe the external conditions necessary for the persistence of the structural self. Differences of the external environment state can therefore relate, through the state transition, to the future x_{t+1} and hence to $\Pi(x_{t+1})$. The paper does not, however, derive non-degenerate dependence on the external environment argument from the type of f alone; which external differences take part in future retention of K is fixed relative to the f, E, E_{req} and state structure of that ACS. The structure in which differences of the external environment state can be reflected in the next state of the environment-response component is described separately from the three principles as the structural environment coupling condition of `subsec:env_coupling`.

Single-time condition versus temporal maintenance (`rem:sm_temporal_scope`). SM_t of `def:SM` is a single-time predicate judged at each $t \in \mathbb{T}_{\mathrm{P},i}$. This paper does not state, as a general consequence of SM_t being true at one time t alone, that $x_s \in K$ holds for later $s > t$. The claim that the state stays in K over a finite interval $\{t, \dots, t+T\}$ ($T \in \mathbb{N}_0$) requires a separate condition on the time sequence: $x_{t+\tau} \in K$ for all $\tau \in \{0, \dots, T\}$. The number of steps of robustness toward the future is described by Hor_f in `subsec:viab_core`, and positive feasibility margin within a finite window by (T1) in `subsec:s3_tstab`; neither is a truth condition of SM_t. To discuss unlimited feasibility preservation, dynamical structure such as state transition, feasible actions, required environmental conditions and feasibility kernels is needed. This paper does not state long-term feasibility as a general consequence of the truth of the single-time SM_t.

Position of this subsection. In Phase 1, the time-indexed satisfaction predicate of SM is positioned, for times $t \in \mathbb{T}_{\mathrm{P},i}$, as the feasibility condition of the current state: SM_t is true $\iff \Pi(x_t) \geq 0 \iff x_t \in K$. This condition is not identified with the existence of a one-step feasible action, the current satisfaction of required environmental factors, blockage, collapse, or the preservation of feasibility over the future. The next subsection introduces the E_{req} robust viability kernel and the finite-step robust feasible sets, and describes, as classification-C quantities, the future-directed side of the principle response item of SM.

References

Clarke, Frank H. 1990. *Optimization and Nonsmooth Analysis*. Vol. 5. Classics in Applied Mathematics. Philadelphia: Society for Industrial; Applied Mathematics (SIAM).

Rockafellar, R. Tyrrell, and Roger J.-B. Wets. 1998. *Variational Analysis*. Vol. 317. Grundlehren Der Mathematischen Wissenschaften. Berlin: Springer.